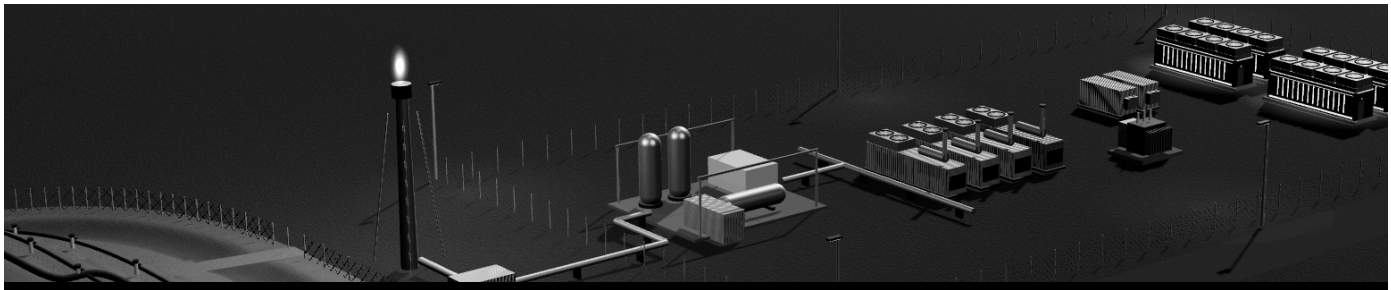
Guadalupe Recycling and Disposal Facility, San Jose, California: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

4.0 MW	4	10 ac	368k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $1.10M	$10k to $20k	$10k to $1.12M



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=1651&CH4=44

THE MATH

Landfill gas collected and flared (2022)	1,651 scfm
Methane content (2022)	44.1%
Energy in the gas	44.2 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 4.0 MW
Compute (power less cooling and losses, /1.2)	about 3.3 MW
Compute modules (about 1 MW each)	4
Gas engines (running plus one spare)	4 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 10 acres
Gas bought per year (95% run time)	368k MMBtu
Gas payment, up to $3.00/MMBtu	up to $1.10M
Land lease at $1,000 to $2,000 per acre per year	$10k to $20k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- This sizing applies only if the owner's own landfill gas power plant is not going ahead
- Latest flow and methane data; closure is now listed for 2044
- Space for a small fenced parcel near the flare
- Gas quality (siloxanes, H2S) and Bay Area air district limits for engines

Source: EPA Landfill Methane Outreach Program (LMOP) database, September 2024 release (2022 data): about 2.38 million cubic feet a day (1,651 scfm) flared at 44.1% methane. The owner's site page (seen 2026-11-04) says the gas is collected and flared, with a permit for a landfill gas power plant under review. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.